Identifying High-Value Locations and Best-Selling Products

An Analysis of Coffee Shop Sales Data

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Table of contents

1 Executive Summary

This report evaluates sales patterns across New York City café locations to identify the strongest site for expansion and the most effective products to prioritise at launch. Although total sales across Hell's Kitchen, Astoria, and Lower Manhattan differ by less than 3%, Hell's Kitchen consistently leads in revenue and shows the most stable demand profile. Coffee is the dominant product category, and items that rank highly in both popularity and revenue represent the strongest candidates for a core launch menu. Based on these findings, Hell's Kitchen is recommended as the optimal location for a new café, supported by a menu centred on high-demand, high-revenue beverages.

2 Introduction

The central problem addressed in this report is how a new café can choose a location and initial product offering that maximises early sales and customer engagement. The report focuses on two key objectives: identifying the location with the strongest potential for expansion, and determining which product should be prioritised at launch. Together, these findings will support strategic decision-making for launching a new café with the highest likelihood of early success.

2.1 Dataset Introduction

To tackle this problem, the analysis uses the Coffee Shop Sales dataset from Maven Analytics, which provides detailed transaction-level information across multiple existing stores. The dataset includes 149116 rows and 11 columns, representing a large volume of customer purchase activity. Key variables include numeric identifiers such as `transaction_id`, `store_id`, and `product_id`, which uniquely identify each transaction, store, and product. Temporal variables include `transaction_date` (stored as character, which is cleaned to Date object later) and `transaction_time` (stored as a time object), capturing when each purchase occurred. Measures of customer behaviour are represented through `transaction_qty`, a numeric variable indicating the number of items purchased per transaction. Categorical variables (stored as character) such as `store_location`, `product_category`, `product_type`, and `product_detail` describe where the transaction took place and the type of product purchased. Together, these variables provide a rich foundation for analysing sales patterns across locations and product categories.

3 Methodology

Before the data is used for analysis, a data cleaning process is performed. All variables are checked for missing values and none are found across the dataset, thus no further action was taken. Next, the class of each variable is verified. The transaction_date variable, which contains date information, was stored as a character string and was therefore converted to a date format to allow for further analysis. Lastly, the data is checked for duplicated rows. As no duplicates are found, all rows are retained for analysis.

Following the goal of determining the best location and best selling products, the relationship between product category, product details, sales quantity, unit price, and store location is explored.

3.1 Summary of data - Sales by location

Table 1: Total sales and quantity sold by store location

Store Location	Quantity Sold	Average Price	Total Sales	Total Transactions	Average Transaction Values	Average Daily Sales
Hell's Kitchen	71737	3.30	236511.2	50735	4.66	1306.69
Astoria	70991	3.27	232243.9	50599	4.59	1283.12
Lower Manhattan	71742	3.21	230057.2	47782	4.81	1271.03

Table 1 presents the total quantity sold, average product price, total sales, total transactions, average transaction values, and average daily sales of stores in each location. The store in Hell's Kitchen recorded the highest total sales amount and highest total transaction count among the three locations, despite having a slightly lower sales quantity compared to the store in Lower Manhattan. Nevertheless, its slightly higher average price contributed to the higher total sales figure.

3.2 Summary of data - Sales by product category

Figure 1 presents the quantity sold by product category. Coffee is the best selling product category, followed by tea. The third best selling category is bakery items; however, the difference between tea and bakery sales is significant, as bakery sold less than half of total tea sold, representing a large gap between the top two best selling categories and the rest.

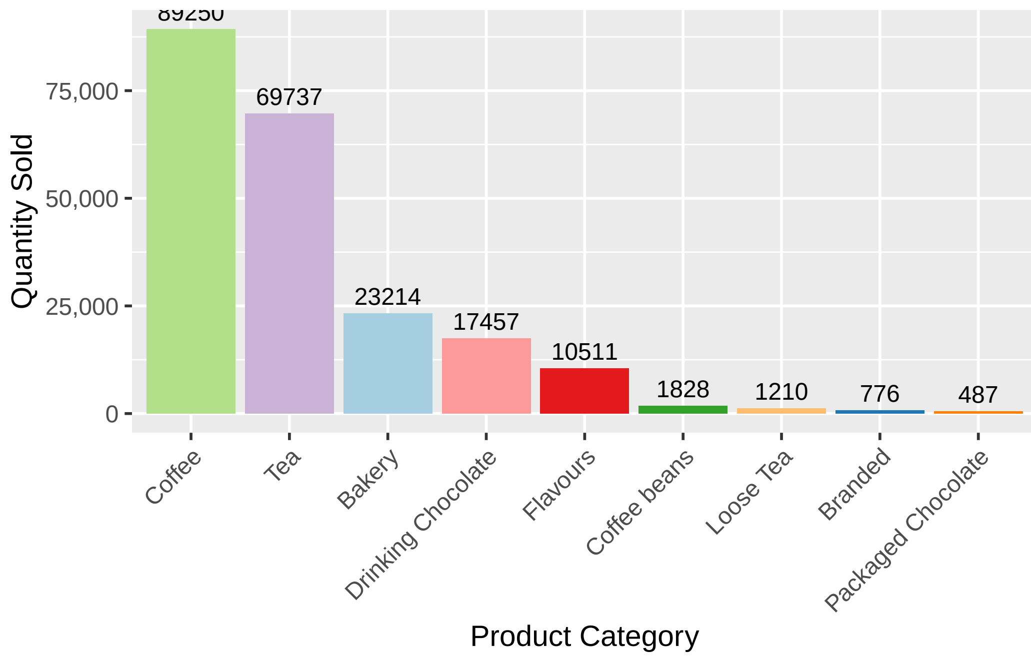

Figure 1: Sales by product category

4 Results

4.1 Which Location would be Most Suitable based on Sales?

4.1.1 Total Sales by Store Location in NYC

Figure 2 shows the total sales across the six-month period are similar across all three locations, with **Hell's Kitchen leading, followed by Astoria and Lower Manhattan**.

4.1.2 Monthly Sales Trends

Figure 3 reveals a **consistent and strong upward trajectory in sales** across all three locations from January to June. All stores dipped slightly in February before recovering and rising gradually to June. This seasonal period likely reflects increased foot traffic in the warmer months, where customers are more likely to leave their homes and visit coffee shops.

Figure 2

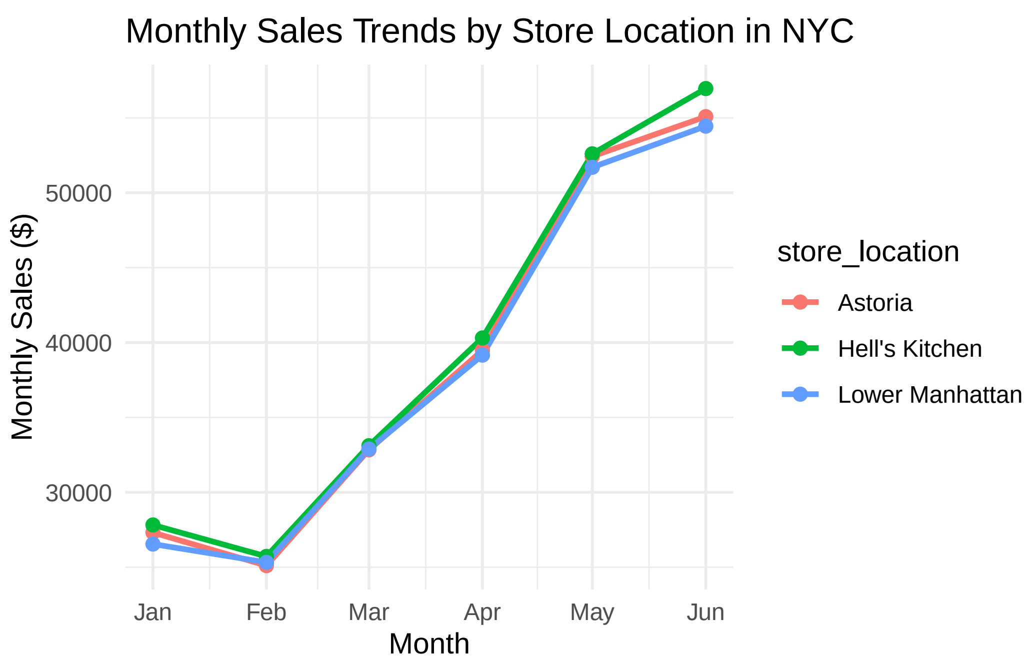

Figure 3

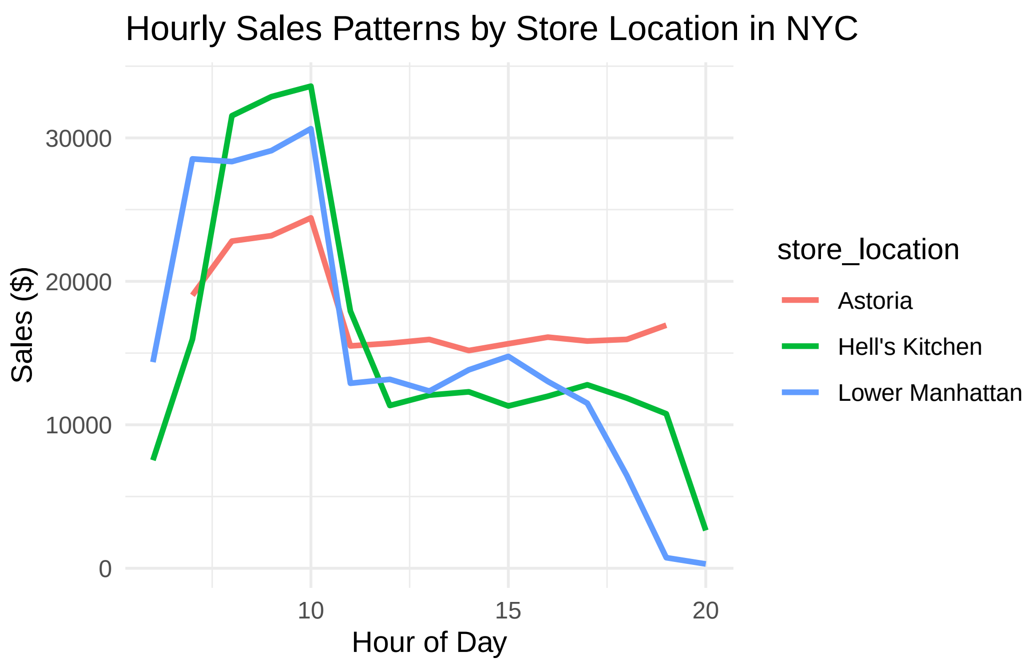

Figure 4

4.1.3 Peak-Hour Demand Analysis

As shown in Figure 4, **all three locations experience the same morning peak**, with sales increasing between 7 to 10 AM. Hell's Kitchen and Lower Manhattan both reached peaks exceeding $30,000 in sales during this time period, aligning with the high foot traffic from morning commuters.

Lower Manhattan drops to nearly $0 by 8 PM, and Hell's Kitchen follows a similar pattern, indicating that these locations are correlated with commuter and office traffic. On the other hand, Astoria sustains sales throughout the evening.

4.2 Optimal Menu Recommendations

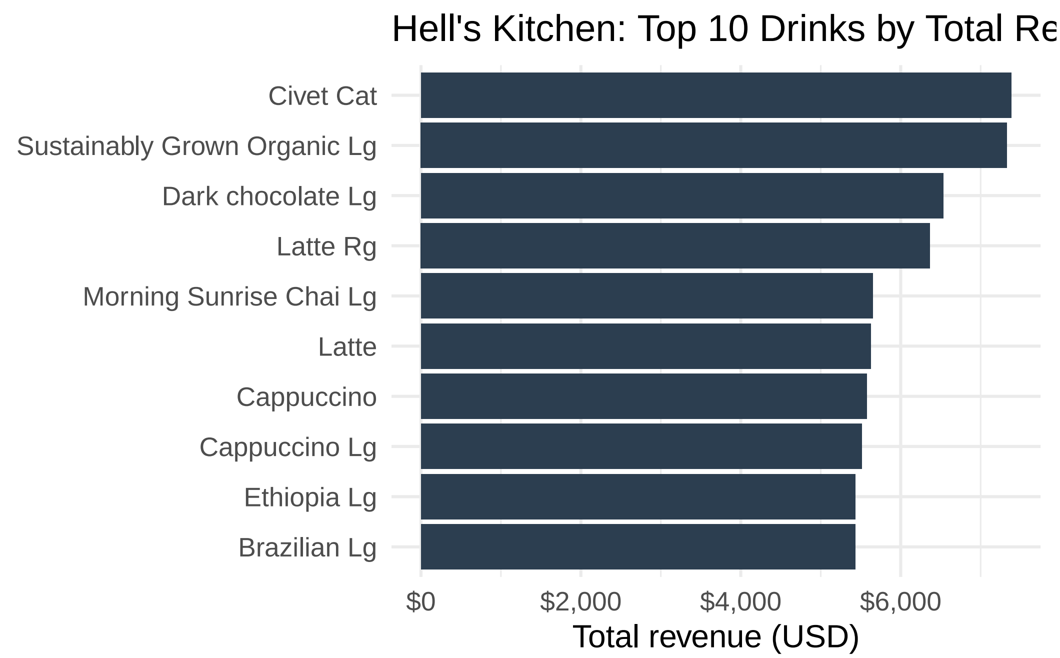

Figure 5: top 10 drinks by total revenue (hell's kitchen)

As shown in Figure 5, total revenue in Hell's Kitchen is concentrated among a small set of drinks, with a clear drop-off after the top-ranked items. The highest-earning products include premium coffees and larger-size beverages, suggesting that both higher unit prices and steady demand contribute to revenue. This figure highlights the key drivers of takings and provides an evidence-based starting point for deciding which drinks should be prioritised for menu placement and promotion.

Table 2: top 10 drinks by total revenue (hell's kitchen)

drink	total_quantity	total_revenue
Civet Cat	164	7380.00
Sustainably Grown Organic Lg	1543	7329.25
Dark chocolate Lg	1452	6534.00
Latte Rg	1498	6366.50
Morning Sunrise Chai Lg	1413	5652.00
Latte	1500	5625.00
Cappuccino	1487	5576.25
Cappuccino Lg	1297	5512.25

Table 2: top 10 drinks by total revenue (hell's kitchen)

drink	total_quantity	total_revenue
Brazilian Lg	1552	5432.00
Ethiopia Lg	1552	5432.00

Table 2 highlights that the Civet Cat generated the highest total revenue in Hell's Kitchen ($7,380), followed by Sustainably Grown Organic Lg ($7,329.25) and Dark chocolate Lg ($6,534). Brazilian Lg and Ethiopia Lg also performed strongly ($5,432 each), suggesting consistent high earners.

Table 3: top 10 drinks by total quantity sold (hell's kitchen)

drink	total_quantity	total_revenue
Ouro Brasileiro shot	1854	5056.20
Serenity Green Tea Rg	1601	4002.50
Morning Sunrise Chai Rg	1589	3972.50
Brazilian Lg	1552	5432.00
Ethiopia Lg	1552	5432.00
Our Old Time Diner Blend Sm	1550	3100.00
Sustainably Grown Organic Lg	1543	7329.25
English Breakfast Lg	1529	4587.00
Earl Grey Rg	1522	3805.00
Brazilian Sm	1520	3344.00

Table 3 shows that the Ouro Brasileiro shot was the most popular item (1,854 units), followed by Serenity Green Tea Rg (1,601) and Morning Sunrise Chai Rg (1,589). Brazilian Lg, Ethiopia Lg, and Sustainably Grown Organic Lg appear as both high-volume sellers and high-revenue items, making them strong Core Menu choices.

5 Discussion

As shown in Figure 5 and Table 2, the revenue in Hell's Kitchen is concentrated among a small set of drinks, led by Civet Cat (bringing in a total of $7,380), Sustainably Grown Organic lg (totalling at $7,329.25), and Dark Chocolate lg (with $6,534 in revenue). However, Table 3 shows that high revenue does not always correspond to high demand: the Ouro Brasileiro shot is the most popular item (1,854 units), while Civet Cat achieves the top revenue rank despite comparatively low volume (164 units), suggesting a premium price point and "specialty" positioning instead.

Taken together, Table 2 and Table 3 support a two-tier menu strategy for launch. "Core menu" items should prioritise drinks that perform strongly on both quantity and revenue, because they provide stable demand and reliable contribution to cafe's the overall revenue; for example, Brazilian lg and Ethiopia lg are high-volume sellers (1,552 units each) and also high revenue earners (both coming in at $5,432 each), while Sustainably Grown Organic lg performs strongly on both metrics (1,543 units sold with a revenue of $7,329.25). In contrast, drinks that rank highly in Table 2 but not in Table 3 (most notably Civet Cat) are best treated as "premium" options to increase the average transaction value through targeted promotion (e.g., staff recommendations or featured placement), while core items are left to anchor consistent day-to-day sales.

6 Conclusion

Tables Table 2 and Table 3 together identify the best menu candidates by combining profitability and demand. Items near the top of both tables are strong "core menu" choices because they sell frequently and generate high total revenue, making them reliable for day-to-day operations. Drinks that rank highly in Table 2 but not Table 3 likely achieve revenue through a higher price point; these are good premium offerings to keep for margin and upselling, but they may require targeted promotion and careful stock planning. Conversely, drinks that are very popular in Table 3 but lower in Table 2 are dependable volume sellers and can be used as anchors for bundles or add-ons. Overall, a balanced menu should prioritise overlapping winners, then supplement with a small set of premium specials.

Furthermore, our analysis also examined the sales performance of Hell's Kitchen, Astoria, and Lower Manhattan, across total sales, transaction behaviour, monthly trends, and hourly patterns. All three locations performed similarly in aggregate, with total sales differing by less than 3% over the six-month period. Monthly trends were consistent across locations, with a dip in February followed by sustained growth through to June, indicating that seasonal demand, rather than location-specific factors, is the dominant driver of revenue fluctuations.

Taking all findings together, **Hell's Kitchen is recommended as the strongest candidate for a new café**. Its more evenly distributed demand and competitive overall sales provide a more stable and sustainable commercial foundation, which are vital for a new café seeking to establish a consistent customer base from the start.

7 Recommendations

- The new cafe should be opened in Hell's Kitchen.
- Profits can be optimised by combining profitability and demand of drinks when choosing menu options.

- Drinks for the menu should be chosen from those which can be identified in Table 2 and Table 3

8 References

Free Sample Dataset Download - Coffee Shop Sales - Maven Analytics | Build Data Skills, Faster. https://mavenanalytics.io/data-playground/coffee-shop-sales. Accessed 21 May 2026.

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